

Problem Statement 1: Student Marks Prediction System

Question: How do you build a Machine Learning model to predict student marks based on study hours and other factors?

Your Task:

Load the dataset using Pandas.

Perform data preprocessing and cleaning.

Select the required features.

Train a Linear Regression model.

Predict the marks for new students.

Evaluate the model performance.

Problem Statement 2: Student Performance Predictor

Question: How can you predict a student's performance index using multiple input features?

Your Task:

Import the Student Performance dataset.

Handle missing values and duplicate records.

Perform feature engineering.

Build a Multiple Linear Regression model.

Predict the Performance Index.

Compare actual and predicted values.

Problem Statement 3: Sales Prediction using Machine Learning

Question: How do you develop a Machine Learning model to predict Walmart sales?

Your Task:

Load the Walmart Sales dataset.

Clean and preprocess the data.

Convert categorical values into numerical format.

Split the dataset into training and testing sets.

Train a Linear Regression model.

Predict future sales and evaluate the model.

Problem Statement 4: Data Preprocessing Pipeline

Question: How do you prepare a real-world dataset before training a Machine Learning model?

Your Task:

Remove missing values and duplicates.

Handle categorical data using Label Encoding and Dummy Variables.

Avoid the Dummy Variable Trap.

Select the important features.

Split the dataset into training and testing data.

Save the processed model using Joblib.

Problem Statement 5: Complete Machine Learning Prediction System

Question: How do you build a complete Machine Learning workflow from data preprocessing to prediction?

Your Task:

Import and explore the dataset.

Perform Exploratory Data Analysis (EDA).

Visualize the data using scatter plots and histograms.

Clean and preprocess the dataset.

Train a Linear Regression or Multiple Linear Regression model.

Test the model with new input data.

Evaluate the accuracy and save the trained model for future use.